

# The evolution of blockchain in smart buildings with IoT integration and future prospects

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## ABSTRACT

The adoption of Internet of Things (IoT) sensors in smart buildings introduces complex challenges around data privacy, integrity, interoperability, and governance. Blockchain—particularly permissioned and hybrid designs—offers immutability, fine-grained access control, and auditable workflows that can strengthen compliance and trust in building operations. This article synthesizes architectural patterns and operational trade-offs for integrating blockchain with smart-building IoT, focusing on latency-aware data placement (off-chain buffering with on-chain anchoring), identity and consent management, and verifiable telemetry for control loops. Across databases (IEEE Xplore, ScienceDirect, Springer, ACM & Wiley), 553 records were identified; Step 1 excluded 311. After duplicate removal and title/keyword screening (Step 2), 375 papers remained with 178 excluded; abstract screening (Step 3) yielded 336 papers with 39 excluded; introduction/conclusion criteria (Step 4) retained 264 with 72 excluded; full-text review (Step 5) resulted in 131 included studies. Consistently with Table 3, counts progressed from 553 (initial) to 375 (title/keyword), 336 (abstract), and 131 (final full-text). We summarize verified design choices and open issues—cross-chain interoperability, light-node feasibility, and empirical benchmarking for off-chain computing—to guide robust, efficient, and modifiable IoT infrastructures.

## 1. Introduction

Smart buildings use IoT sensors to manage lighting, HVAC, security, and other services, improving efficiency and comfort, yet they face persistent issues in security, privacy, interoperability, and scalability. Blockchain, as a decentralized and tamper-evident ledger, is a promising enabler for trustworthy automation and auditability. Fig. 1 illustrates the integration of Blockchain and IoT in smart buildings.

Following the advent of cryptocurrencies, Bitcoin demonstrated decentralized consensus at scale and catalyzed the smart-contract ecosystem through Ethereum [1–3]. Blockchain records transactions

immutably across distributed nodes, enabling transparent, auditable interactions among IoT devices, building systems, and users while reducing reliance on a central authority [4,5]. The synergy with AI further supports secure data sharing, automated execution, and privacy-preserving analytics across domains such as healthcare and insurance [6–8], fostering open, decentralized applications and real-time, cross-system coordination in smart buildings [9–12]. Sensors supply real-time data for control and monitoring but introduce operational, privacy, and security risks, including network attacks and maintenance complexity [13–16], alongside data-management, accuracy, interoperability, and scalability challenges in diverse environments [17,18].

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Blockchain mitigates these risks by distributing trust, enforcing integrity via cryptographic hashing and consensus, and enabling controlled, role-based access to sensitive telemetry [19–22].

Key challenges: security and privacy of occupant-linked telemetry; interoperability across heterogeneous devices/protocols; scalability and latency for real-time control; and auditable data governance across stakeholders.

Why blockchain is important in smart buildings: It provides a secure, transparent, and decentralized foundation for high-volume, occupant-linked telemetry; removes single points of failure; and creates immutable, auditable records that build stakeholder trust. Via smart contracts, it automates device authentication and access control and offers a common, verifiable layer for interoperability across heterogeneous systems—thereby improving resilience, governance, and end-to-end integrity in real-time building operations [4,5,19–22].

Main contributions of this paper:

- A compact synthesis linking these challenges to permissioned/hybrid blockchain patterns for smart buildings;
- A verification-oriented view (on-chain metering, selective-disclosure, oracle design) for trustworthy operations without exposing raw identities;
- Practical guidance on latency-aware data placement (off-chain buffering with on-chain anchoring) and governance fit;
- Open issues to steer future work (cross-chain interoperability, oracle robustness, empirical benchmarks for light nodes and off-chain computing).

## 2. Background and fundamentals of blockchain–IoT in smart buildings

### 2.1. Current research

The use of IoT sensors in smart buildings has emerged as a significant development in building management systems today. IoT sensors are essential components for automating various activities within a building, including energy management, lighting, HVAC systems, and

security surveillance. These sensors gather real-time information on the environment, including temperature, humidity, occupancy, and air quality. This information can help building managers improve energy utilisation, security, and enhance comfort for occupants [23]. In other words, filters such as occupancy sensors can help lower energy consumption by adjusting lighting, heating, ventilation, and air conditioning systems when rooms are unoccupied, thereby making the consumption more efficient and cost-effective [24]. Additionally, the environmental sensors monitor temperature and air quality, ensuring occupants experience a comfortable and healthy environment through the functioning of the HVAC system. Recent surveys and application studies also underline the centrality of sensor-driven automation to data integrity and auditability when combined with blockchain anchoring, particularly for compliance and traceability in building operations [25–28].

Nevertheless, the implementation and control of IoT sensors in smart buildings present several challenges. The primary concern is data privacy and security, as a substantial volume of data collected with the aid of IoT sensors may be vulnerable to cyberattacks, including data breaches, hacking, or unauthorised access [29]. Additionally, interoperability is also a problem because devices manufactured by different developers may lack the ability to interact with one another due to variations in communication protocols. This restricts the smooth incorporation of IoT systems and poses weaknesses to the smart building infrastructure [30]. Blockchain technology has become a promising solution to address the challenges faced by IoT systems in smart buildings. Blockchain can thereby guarantee the sovereignty and privacy of data shared among IoT devices by offering a secure, transparent, and unforgeable ledger. What the decentralisation of blockchain eliminates is the presence of a centralised authority to control the data, consequently improving security considerably and virtually eliminating any risk of single points of failure [31]. Very recent reviews (2023–2025) corroborate this trajectory, reporting stronger privacy guarantees and tamper-evident logging when IoT telemetry is anchored to permissioned ledgers or hybrid on/off-chain designs, and highlighting remaining gaps in interoperability and governance [26–28].

This technology provides energy and cost optimisation solutions by

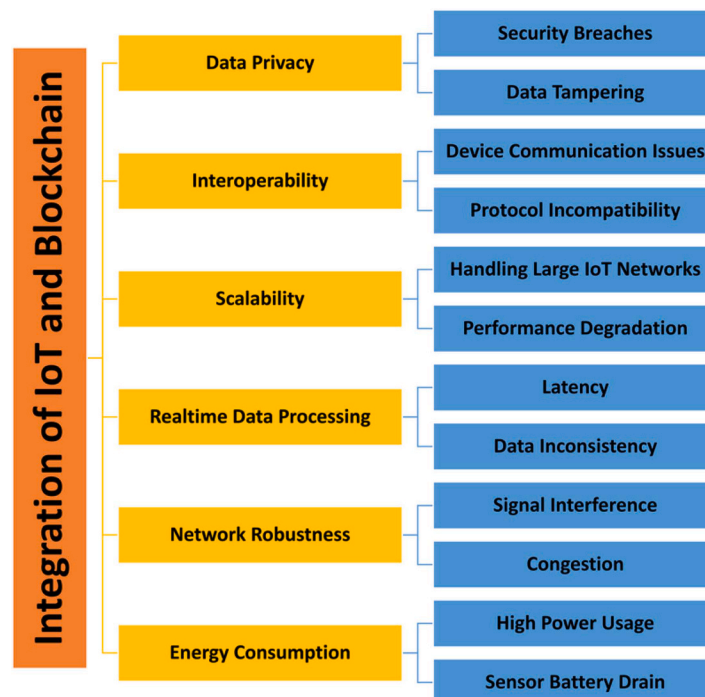


Fig. 1. Overview of blockchain and IoT integration in smart buildings.

analysing the daily and weekly electricity usage of building appliances using real-time data. Smart buildings, including residential and commercial structures, can leverage IoT and 5 G improved mobile broadband, as previously indicated [32]. The authors of this paper discuss concerns over the trends, present the advantages and disadvantages, and outline the challenges associated with IoT implementation in built-up areas. In [33], the authors highlight five key aspects, considerations, and challenges of IoT-oriented blockchains. A robust security blockchain platform for IoT faces various challenges, according to the authors. The researchers studied two blockchain topologies and found that the device-to-device architecture has higher throughput than gateway-based topologies. In the study [34], the structure of smart cities and the related concerns regarding data protection and privacy are summarized. The qualities, applications, and challenges of blockchain technology, including the utilization of smart cities, are also addressed. Complementing these works, recent surveys emphasize Layer-2 scaling (channels/rollups), sidechains, and off-chain computation to meet building-scale throughput and latency demands without sacrificing auditability [27,28]. Application-oriented studies in energy and grid-interactive buildings also document blockchain's role in transactive energy, DR settlement, and secure metering [35].

In another study [36], to accommodate the decentralized architecture of blockchain technology, the authors suggested sophisticated authentication modifications. Additionally, there was a reasonable discussion regarding security concerns associated with authenticating smart devices with IoT devices. They critique and analyze the literature evaluation of models that employ authority protocols to evaluate computational resources, communication overheads, and system structure for resilience. The study by the authors [37] thoroughly examines the introduction of blockchain-based IoT security, focusing on its shortcomings, development, and general aspects of the embedded security structure. One of the most important and promising trends in computational communication systems is the collaboration between Blockchain and IoT. In another study [38], the authors explore key factors in integrating smart building AI and blockchain. The authors suggested ways to improve blockchain security and summarized key elements that could assist in constructing intelligent transport systems based on Blockchain-AI. Table 1 presents a comparative analysis of existing surveys on Blockchain, IoT, and Smart Cities. With the help of the expanding Internet of Things, smart devices produce vast volumes of data to support multiple edge services. Making sure that aggregated data, which may contain sensitive information, is secure and private is a basic challenge in the Internet of Things (IoT) [39]. In the era of Industry 4.0, the Industrial Internet of Things (IIoT) offers a lot of potential. However, there are significant privacy and security issues with IIoT industrial applications that collect and track data independently and in real time. User authentication is a popular and efficient method of determining a user's level of trustworthiness, which affects data security and privacy in IIoT applications [40].

Newer contributions extend these security perspectives with end-to-end device authentication, access control, and automated policy enforcement via smart contracts, demonstrating improved resistance to spoofing and configuration drift in building services [26,41,42]. The confluence of AI/ML with blockchain is also increasingly reported to enhance anomaly detection, predictive maintenance, and autonomous control loops in building systems, provided that data minimization and explainability considerations are met [26,27,43]. Table 1 shows the comparative study of existing surveys on Blockchain, IoT, and Smart City.

Security and privacy have been the two primary concerns in the IoT arena, particularly in the smart building environment, where data on occupants and building systems is constantly generated and transmitted. It has been established that IoT devices are vulnerable to various security threats, including data interception, unauthorized access, and denial-of-service attacks [60]. With the massive introduction of IoT sensors into smart buildings, the attack surface available to

**Table 1**

Comparative study of existing surveys on blockchain, IoT, and smart city.

| Topic               | Main Focus                                  | Key Contributions                                                                   | References |
|---------------------|---------------------------------------------|-------------------------------------------------------------------------------------|------------|
| Blockchain          | Blockchain architecture & security          | Forking, cryptography, networking, consensus, privacy, challenges, and applications | [41]       |
|                     | Blockchain security risks                   | Security attacks, solutions, and future directions                                  | [42]       |
|                     | Cross-domain survey of blockchain for IoT   | Taxonomy, integration patterns, open issues (interoperability, governance)          | [27,28]    |
| Smart Contract      | Smart contract lifecycle                    | Operating mechanism, framework, challenges, and trends                              | [43]       |
|                     | Security vulnerabilities in smart contracts | Identification, exploitation, and mitigation                                        | [44]       |
| Smart Contract      | Policy automation & access control          | Smart-contract enforcement in IoT/BIoT; oracle patterns; governance                 | [26,42]    |
| Smart City          | Privacy and security in smart cities        | Architecture, privacy issues, security requirements, and solutions                  | [45]       |
|                     | Data-centric approach                       | Data flow, technologies, security, and privacy                                      | [46,47]    |
|                     | Privacy-first design                        | Privacy taxonomy, attacks, and protection methods                                   | [48]       |
| (IoT)               | Cybersecurity and privacy                   | Requirements, enabling technologies, and future directions                          | [49]       |
|                     | ML + Blockchain for smart cities            | Comprehensive survey on ML-enabled Blockchain/IoT in smart-city services            | [50]       |
|                     | IoT security and challenges                 | IoT architecture, threats, and blockchain-based mitigation                          | [51]       |
| Blockchain + IoT    | IoT-Blockchain convergence                  | Security models, edge/off-chain patterns, standardization outlook                   | [26,41]    |
|                     | Integration challenges                      | Blockchain-IoT integration, benefits, and platforms                                 | [52]       |
|                     | Decentralized IoT systems                   | Privacy, trustless architecture, identity, and monetization                         | [53]       |
| Blockchain + IIoT   | BIoT applications                           | Optimized blockchain design and challenges                                          | [50]       |
|                     | Threat models and privacy                   | Classification of attacks, research challenges                                      | [54]       |
|                     | State of the art & future directions        | Layer-2 scaling, hybrid on/off-chain storage, interoperability frameworks           | [25–28]    |
| Blockchain + IIoT   | Blockchain in industrial IoT                | Trends, application areas, and open issues                                          | [55]       |
|                     | Security enabler in IIoT                    | Blockchain applications to enhance IIoT security                                    | [56]       |
|                     | Blockchain roles in IIoT                    | As a trusted party, security platform, access control, and payments                 | [57]       |
| Blockchain + Smart  | Industry 4.0/5.0 implementations            | Authentication, access control, performance constraints in IIoT                     | [41,42]    |
|                     | Blockchain in smart cities                  | Applications, challenges, research scope, and ICT enhancement                       | [58]       |
|                     | Smart city improvement via blockchain       | Review of smart city challenges and blockchain-based solutions                      | [59]       |
| Energy / Smart-grid | Grid-interactive buildings & energy         | Transactive energy, secure metering, DR settlement with blockchain                  | [35]       |
| Our Work            | Blockchain-based smart cities               | Blockchain evolution, platforms, case studies, data needs, and new challenges       |            |

cybercriminals grows, making it more challenging to secure sensitive information. While the present study concentrates on architectural integration, future empirical validation will benchmark the misidentification rate of PUF-based device identity binding under diverse operating conditions and evaluate the memory and energy footprint of post-quantum cryptographic algorithms on embedded MCUs. These measurements will quantify reliability and resource trade-offs, ensuring that the proposed blockchain-IoT framework remains both lightweight and quantum-resilient in real-world deployments. The majority of recent research on decentralized Internet of Things apps concentrates on particular weaknesses, and there are not many methods specifically built to address privacy and trust issues. Existing issues are being addressed by the quick adoption of blockchain-based solutions to increase the security and reliability of Internet of Things networks. Device identity storage and validation can be done decentralized with blockchain-based authentication systems. This enables safe and confidential communication between devices as well as between devices and other systems [61].

Blockchain offers an effective cure to these security problems. Its architecture is decentralized; thus, the data is spread across multiple nodes, which makes centralized attacks risky. In addition, blockchain utilizes the latest cryptographic infrastructure to safeguard the privacy of the information being shared, and only authorized users can access it. Unaddressed security and privacy issues have been studied through several studies, with specific considerations on how blockchain can address these problems. For example, some studies have suggested using blockchain as a source of authentication, effectively allowing only trusted devices to communicate with IoT sensors and thereby mitigating the threat of malicious attacks [62]. Further studies have focused on utilizing blockchain as a mechanism to encrypt information in a manner that maintains confidentiality and security during transmission [63].

Blockchain solutions have proven effective in mitigating several issues experienced by IoT systems in smart buildings, including data privacy, scalability, interoperability, and real-time data processing. It has been identified as one of the most promising opportunities for using blockchain in smart buildings, as it enables the creation of a safe and efficient data management system. Due to the blockchain storage of IoT sensor data, the data will be immutable, meaning it cannot be changed once logged. This eliminates the chances of tampering or manipulation, as the information used in decision-making is accurate. When everyone is satisfied with the data, there can be no room for manipulation [24].

Moreover, blockchain has the potential to boost scalability through the decentralization of data. In the case of large-scale smart buildings, where numerous IoT devices are implemented, it may be convenient to control data within a single centralized system. However, due to the potential load on the system, bottlenecks and performance problems may occur. Blockchain can disseminate information to numerous servers and nodes, allowing the system to scale more effectively without compromising performance [64]. Additionally, blockchain enables different IoT devices/products developed by various manufacturers to interoperate, allowing them to communicate and share data seamlessly without compatibility issues. This is essential for the successful integration of different IoT devices in smart buildings, enabling them to collaborate and coordinate the optimal functioning of the buildings.

## 2.2. Problem statement and scope

Smart-building IoT deployments must satisfy five concurrent requirements that current practice does not jointly meet: (i) privacy-preserving, tamper-evident telemetry; (ii) interoperable data exchange across heterogeneous devices and protocols; (iii) scalable throughput with low, predictable latency suitable for control loops; (iv) robust identity, access control, and auditable governance; and (v) cost-aware, maintainable operation in real buildings. The literature consistently documents privacy and security risks from large-scale sensor data collection in buildings [29,39–42,45,48–51,60–62], protocol and platform fragmentation that hinders end-to-end interoperability [30,26–28,

41,52–57], and performance constraints when naïvely centralizing high-rate telemetry [25–28,33,41,64]. While blockchain (especially permissioned and hybrid on/off-chain patterns) offers immutability, fine-grained access, and verifiable logging [25–28,31,35,41–43,50], recent surveys emphasize that throughput, latency, and cross-domain interoperability remain open issues without careful architectural choices (e.g., sidechains/Layer-2 and off-chain computation) [25–28,41,42].

Accordingly, the problem addressed in this work is to identify and justify architecture patterns that:

- protect data confidentiality and integrity while supporting auditability and consent management for sensor streams [25–28,31,41–43];
- enable interoperable operation across vendor ecosystems and building subsystems via standardized interfaces and policy automation (smart contracts/oracles) [26,27,42,43,50,52–54];
- achieve building-scale scalability—throughput and confirmation latency compatible with operational decision cycles—using hybrid storage (off-chain buffering with on-chain anchoring), Layer-2/sidechains, and event-driven pipelines [25–28,33,41]; and
- provide device identity and access-control primitives suited to constrained hardware (e.g., lightweight authentication, key management) with auditable governance [39–42,60–62].

In this work, we focus on permissioned and hybrid blockchain designs integrated with IoT data pipelines in residential and commercial buildings, considering energy/HVAC, lighting, and security telemetry. We do not propose new cryptographic schemes; instead, we synthesize deployment-ready patterns for identity, access control, storage placement, and governance, and report their trade-offs under the constraints surfaced by recent reviews and application studies [25–28,31,33,35,41–43,50,52–57,64]. This scope directly reflects the gaps highlighted across the reviewed works—privacy and trust at scale, standards-aligned interoperability, and empirical performance baselines for hybrid on/off-chain integration—and it guides the architectural design recommendations and discussion developed in the remainder of the paper.

## 3. Methodology

This work develops a precise SLR to analyze real-world evidence on IoT-enabled smart buildings and blockchain integration, aiming to address challenges in smart sensor systems within these buildings. Smart buildings are crucial and one of many IoT and Blockchain domains explored in this research. This SLR examines how people have linked IoT sensors with blockchain and overcome obstacles. This study also discusses the challenges of smart building IoT sensors and how blockchain addresses them. This will be achieved by exploiting and defending research questions in the review process. Fig. 2 illustrates the methodological planning process, while Table 2 outlines the research topics. In addition, and to address performance concerns raised by reviewers, we explicitly incorporate a lightweight consensus latency-throughput quantification in the methodology to make concurrency bottlenecks and edge-side delays measurable (see Section 3.6).

Table 2 outlines the research questions and objectives related to IoT challenges, blockchain solutions, and their integration in smart buildings. It aims to explore issues such as data security, scalability, and the potential applications of blockchain in IoT systems.

### 3.1. Select data sources

Library and repository sources for other search results are referred to as actual sources. Springer Digital Library, ACM Digital Library, IEEE Xplore, and ScienceDirect are our top digital libraries for results. To accurately interpret past findings, access to full-text literature is necessary. Any digital library has several ways to find information. To find





Fig. 2. Proposed systematic literature review (SLR).

**Table 2**

Research questions and objectives.

| Research Questions                                                                                               | Research Objectives                                                                                                      |
|------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| 1. What are the main challenges faced by IoT sensors in smart buildings?                                         | 1. To identify and analyze the key challenges faced by IoT sensors in smart buildings.                                   |
| 2. How can blockchain technology address data privacy, security, and interoperability issues in smart buildings? | 2. To explore the potential solutions provided by blockchain technology in addressing these challenges.                  |
| 3. What are the potential applications of blockchain and IoT integration in smart buildings?                     | 3. To evaluate the benefits of integrating blockchain with IoT systems in terms of security, efficiency, and automation. |
| 4. How can blockchain enhance the scalability and performance of IoT systems in smart buildings?                 | 4. To investigate prospects and advancements in blockchain technologies for smart buildings.                             |

reliable information, research methods must be adapted to align with resource goals. Table 3 and Fig. 3 present the data sources and research derived from search queries.

The table provides a detailed breakdown of the scientific research papers submitted to various publishers, including ACM, IEEE, ScienceDirect, Springer, and Wiley. It categorizes the data into four stages: submissions, reviews, accepted papers, and rejected papers. The numbers reflect the progression of each submission through the review and acceptance process. In total, 131 papers have successfully passed through the review process and have been finalized for publication. This analysis highlights the trends in submission and acceptance across various research platforms.

### 3.2. Formulate search string

To answer research inquiries, keywords and search terms related to the study are merged into a search string. Table 4 shows how research question-related phrases are used to structure search strings. The supplied keywords are used in the order specified using 'AND' and 'OR' logical operators to generate each search string.

**Table 3**

Query results from data sources.

| Library        | Initial | Title and Keyword | Abstract | Introduction and Conclusion |
|----------------|---------|-------------------|----------|-----------------------------|
| ACM            | 60      | 45                | 36       | 20                          |
| IEEE           | 150     | 95                | 85       | 30                          |
| Science Direct | 122     | 85                | 75       | 26                          |
| Springer       | 131     | 90                | 90       | 40                          |
| Wiley          | 90      | 60                | 50       | 15                          |
| Results        | 553     | 375               | 336      | 131                         |

Fig. 4 depicts the detailed process of formulating the search string, outlining the steps involved in selecting keywords, applying Boolean operators, and refining search queries to gather relevant literature for the study.

### 3.3. Define inclusion and exclusion criteria

The comprehensive literature review will encompass research that meets the criteria mentioned above. Selection is based on the following parameters: All research must be in English and have been published between 2017 and 2025. Research should focus on integrating IoT sensors in smart buildings with blockchain technology to address specific problems. The study examines the integration of IoT sensors in smart buildings utilizing blockchain technology. All selected papers must assess IoT sensors, smart buildings, their factors, and the limitations of IoT sensors, as well as explain why blockchain is needed to overcome these limitations. Consider only publications from respected journals, books, and conferences. No studies meet the exclusion criteria for the literature review. All pre-2017 papers are excluded.

### 3.4. Define quality assessment criteria

Qualitative methods assess research quality, credibility, and worth in systematic literature reviews, SLRs, and meta-analyses. Qualitative methods are used to determine the validity and structure of quality claims. Quality control and confirmation of findings help ensure that our main investigations deliver enough data to answer our research objectives. This stage compares each study topic to the standard. Table 5 shows the quality control system from QCS.

### 3.5. Primary study selection

The phrase "primary study" refers to book sections and subtopics that answer a research issue. To efficiently complete this study, relevant papers were identified utilizing the five-stage tollgate approach. 116 original studies were discovered using pre-set quality criteria. Fig. 5 and Table 6 show the final paper selection, while Fig. 6 illustrates the prisma diagram.

Fig. 5 displays the final paper selection based on year-wise distribution, showing the number of papers selected from 2017 to 2025.

Fig. 6 illustrates the systematic process of paper selection across multiple databases, including IEEE Xplore, ScienceDirect, Springer, and ACM & Wiley. Initially, 553 papers were identified through keyword searches, with 311 excluded in Step 1. In Step 2, duplicates were removed, and further inclusion/exclusion was applied by reviewing titles and keywords, narrowing the selection to 375 papers, with 178 excluded. Step 3 involved reading abstracts, reducing the number to 336 papers, with 39 excluded. Step 4 applied inclusion and exclusion criteria

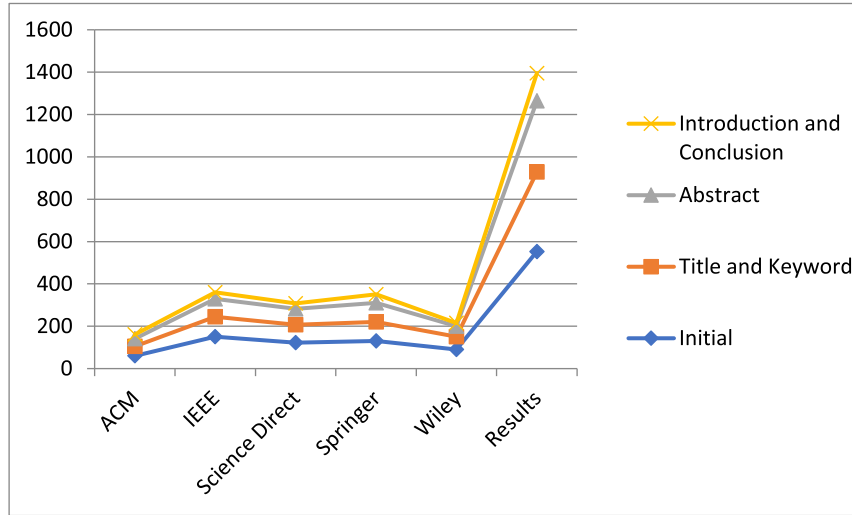


Fig. 3. Search queries representation.

**Table 4**  
Search string formulation.

| Keyword                   | Synonym / Alternative word                                                      |
|---------------------------|---------------------------------------------------------------------------------|
| Blockchain                | ("Distributed Ledger," "DLT," "Blockchain Technology")                          |
| Internet of Things        | ("IoT," "Internet of Things," "Connected Devices")                              |
| Smart Buildings           | ("Intelligent Buildings," "Automated Buildings," "Building Management Systems") |
| Data Privacy              | ("Data Protection," "Confidentiality," "Data Security")                         |
| Interoperability          | ("Device Communication," "System Compatibility," "Protocol Standardization")    |
| Scalability               | ("System Expansion," "Network Growth," "Performance Scaling")                   |
| Real-time Data Processing | ("Data Streaming," "Instant Data Processing," "Low-latency Processing")         |
| Security                  | ("Cybersecurity," "Data Integrity," "Access Control")                           |
| Energy Efficiency         | ("Energy Optimization," "Energy Management," "Sustainable Energy Use")          |
| Smart Contracts           | ("Self-executing Contracts," "Automated Contracts," "Blockchain Contracts")     |
| Automation                | ("Autonomous Systems," "Automated Management," "Self-regulating Systems")       |
| Blockchain Integration    | ("Decentralized Integration," "Distributed Ledger Integration")                 |

based on the introduction and conclusion, leaving 264 papers, with 72 excluded. Finally, in Step 5, the full text of 131 papers was reviewed, resulting in the final selection for the study.

### 3.6. Latency-throughput quantification for lightweight (Edge) consensus

To rigorously address high-concurrency IoT streams versus blockchain throughput, we distinguish the aggregate IoT arrival rate  $\lambda_{IoT}$  (events/transactions per second) from the effective blockchain service rate  $\mu_{BC}$  (confirmed transactions per second). A necessary feasibility condition for bounded delay is  $\lambda_{IoT} \leq \mu_{BC}$ ; if  $\lambda_{IoT} > \mu_{BC}$ , queues grow at ingress, signaling a throughput bottleneck. For edge-assisted lightweight consensus, we decompose end-to-end latency into additive components:

$$L_{e2e} = L_{ingress} + L_{consensus} + L_{commit} \quad (1)$$

Here,  $L_{ingress}$  captures local buffering and pre-validation at gateways,  $L_{consensus}$  accounts for leader election/voting and message rounds of the chosen lightweight protocol, and  $L_{commit}$  reflects block finalization or periodic anchoring (e.g., sidechain-to-main-chain). For batched designs with block size  $b$  and nominal inter-block interval  $\Delta$ , an operational upper bound on service capacity is  $\mu_{BC} \leq b/\Delta$ , subject to protocol

overheads and invalidation rates. Practically, deployments should surface the tuple  $(L_{ingress}, L_{consensus}, L_{commit})$ , together with  $\lambda_{IoT}$  and target  $\mu_{BC}$ , to monitor concurrency pressure and ensure building control loops remain within timing constraints under bursty loads.

## 4. Results and discussion

Resource, energy, and security management in smart buildings has become a revolution through the incorporation of IoT sensors. Nonetheless, some challenges emerge when IoT sensors are integrated, affecting effectiveness and efficiency levels. The following are the critical challenges in IoT sensor systems for smart buildings.

### 4.1. Privacy and security

The IoT sensors gather large quantities of information on an ongoing basis, some of which may be confidential (occupancy or temperature preferences, personal behavior patterns, etc.). Such exposure of the data can pose serious privacy concerns for the occupants of buildings. The problem is further exacerbated by the fact that smart buildings may be equipped with numerous IoT devices [65], each with its own security protocols and standards to follow. The inadequacy of sound security models also increases the likelihood of unauthorized access to data, thereby exposing privacy weak points [66]. In this context, blockchain-based authentication and audit trails (discussed below) specifically target confidentiality and integrity risks tied to multi-vendor device fleets.

### 4.2. Blockchain-enabled mitigations and solution mapping

Table 7 synthesizes the key relationships between the challenges encountered in IoT-enabled smart buildings and the corresponding blockchain-based mitigation strategies. It demonstrates that decentralized authentication and smart-contract governance directly strengthen data privacy, while cross-chain gateways and standard models enhance interoperability among heterogeneous devices. Hybrid architectures and lightweight consensus mechanisms improve scalability and latency performance, whereas immutable ledgers ensure verifiable audit trails for all operational data. Finally, oracle-driven validation and consortium governance frameworks provide the trust and accountability needed for regulatory compliance. Despite these benefits, certain trade-offs—such as computational overhead, synchronization delay, and administrative complexity—remain, underscoring the need for adaptive, context-aware blockchain designs that balance transparency, efficiency, and privacy in

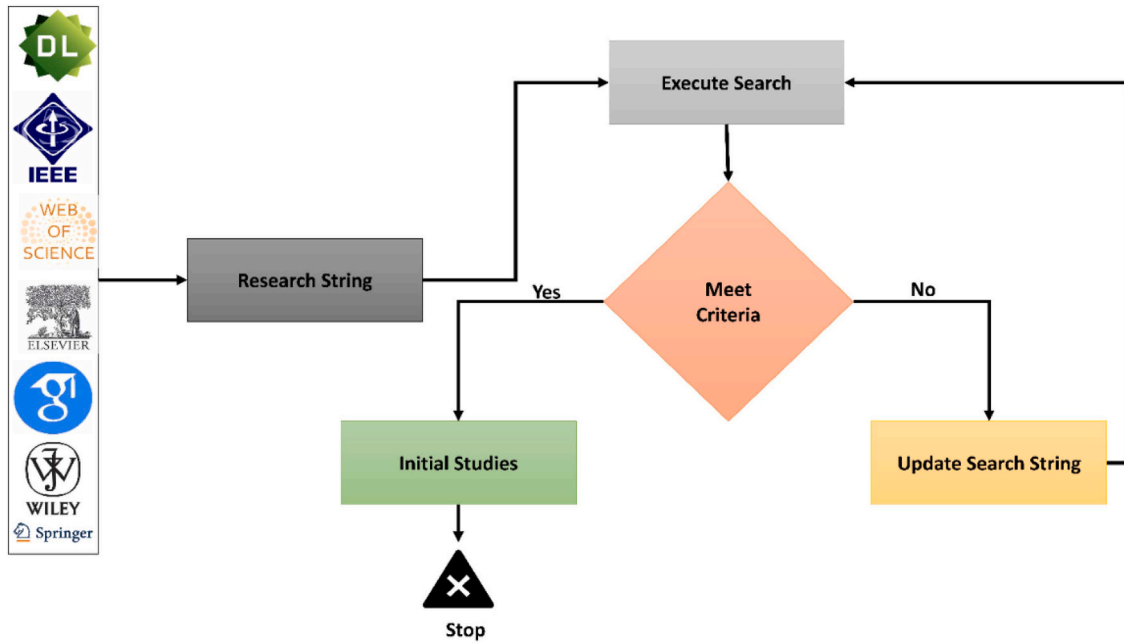


Fig. 4. Process of formulating search string.

**Table 5**  
Quality assessment criteria.

| Sr. No | QA Questions                                                                                                                             |
|--------|------------------------------------------------------------------------------------------------------------------------------------------|
| QCS1   | Does the study provide enough information on the importance of IoT sensors in smart buildings?                                           |
| QCS2   | Does the study address the challenges that IoT sensors face in smart buildings, particularly in terms of security and other key aspects? |
| QCS3   | Does the study offer effective blockchain-based solutions to the challenges faced by IoT sensors in smart buildings?                     |

smart-building ecosystems.

#### 4.3. Interoperability across heterogeneous systems

One of the most significant challenges in IoT systems is interoperability, which refers to the ability of devices manufactured by different companies to interact seamlessly in various systems. The IoT has sensors of different vendors installed in smart buildings, and they may employ multiple communication protocols. This inconsistency makes the devices inefficient in sharing data and interacting effectively. When this

happens, the system tends to be inadequate and also becomes more susceptible. The issue of interoperability is especially relevant to smart building ecosystems, where a variety of systems (lighting, HVAC, security, energy control, etc.) must be integrated to achieve maximum efficiency [67].

#### 4.4. Scalability under growing device density

The scalability of IoT sensor networks is a critical issue as smart buildings become larger and more complex. Due to the increasing number of sensors and IoT devices being added to large-scale buildings, a stronger infrastructure is necessary to support the data being generated. The latter problem is exacerbated by the fact that smart building systems are growing in size and complexity to accommodate new services and technologies, making this scalability issue even more fundamental. In the absence of scalability mechanisms, IoT networks may be expected to perform at lower levels, resulting in delays, non-optimality, or even system crashes [32].

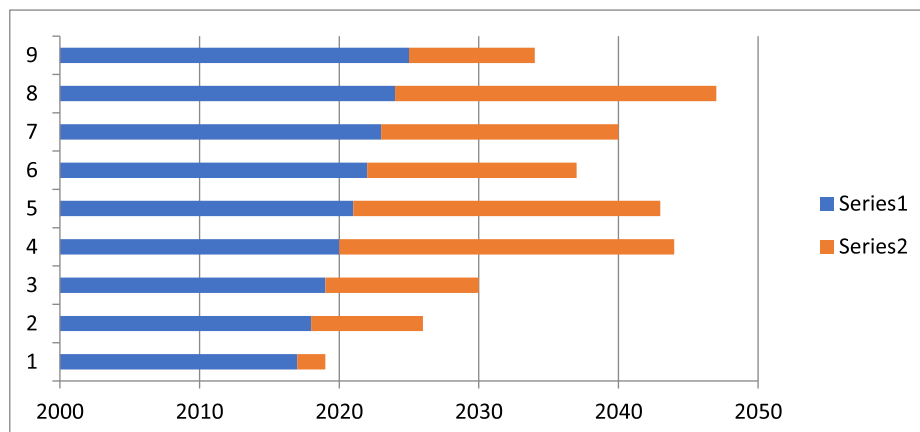


Fig. 5. Final paper selection.

**Table 6**  
Final paper selection.

| Year | Total # Papers |
|------|----------------|
| 2017 | 2              |
| 2018 | 8              |
| 2019 | 11             |
| 2020 | 24             |
| 2021 | 22             |
| 2022 | 15             |
| 2023 | 17             |
| 2024 | 23             |
| 2025 | 9              |

#### 4.5. Real-time data processing requirements

Every day, data produced by IoT sensors should also be processed in real time and be subject to immediate action to maximize building operations. The volume, velocity, and variety of data produced by such sensors pose a significant challenge, as they require real-time data processing [68]. Problems with latency, inconsistent data, and the time lag between data and action may become obstacles to decision-making and the overall effectiveness of building management systems [69]. It is therefore essential that data processing be accurate and efficient, enabling energy optimization while maintaining security and occupant comfort.

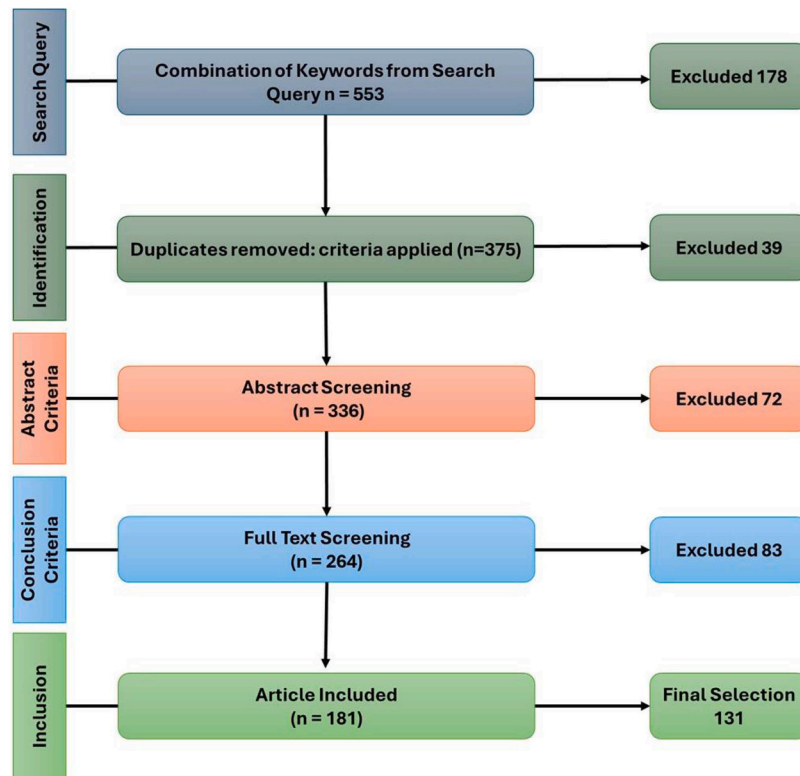


Fig. 6. PRISMA diagram.

**Table 7**  
Mapping of IoT challenges in smart buildings to blockchain-based solutions.

| IoT Challenge                                     | Blockchain-Based Solution                                      | Mechanism / Implementation                                                                                         | Example or Case Study                                                                     | Limitations / Trade-offs                                                           |
|---------------------------------------------------|----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Data privacy and security                         | Decentralized authentication and smart-contract access control | Device identities verified through permissioned ledgers; smart contracts enforce selective access and audit trails | Secure energy-metering or occupancy-tracking systems using on-chain identity management   | Increased on-chain cost; potential latency in high-frequency authentication events |
| Interoperability across heterogeneous IoT devices | Cross-chain bridges and standard data models                   | Blockchain gateways enable data exchange among different IoT protocols; APIs unify device interaction formats      | Multi-building management systems integrating Modbus and BACnet via blockchain middleware | Added protocol translation overhead; risk of bridge vulnerabilities                |
| Scalability and latency in real-time control      | Layer-2 and hybrid on/off-chain architectures                  | Lightweight consensus (e.g., PBFT) and off-chain computation for delay-sensitive transactions                      | Hybrid blockchain-edge frameworks in energy trading and HVAC control                      | Synchronization delays; complexity of data consistency between chains              |
| Data integrity and auditability                   | Immutable ledger with cryptographic hashing and timestamping   | Transactions hashed and chained to prevent tampering and ensure traceability                                       | Building occupancy data and sensor readings stored with tamper-evident verification       | Storage growth over time; high throughput demands for continuous data logging      |
| Trust management and accountability               | Reputation and oracle-based validation                         | Oracles feed verified external data (e.g., pricing, occupancy) into smart contracts for automated actions          | Verified service contracts for maintenance and security alerts                            | Oracle reliability and trust anchoring remain challenges                           |
| Governance and compliance                         | Consortium and permissioned blockchain frameworks              | Stakeholders (owners, tenants, service providers) share nodes under role-based access governance                   | Building energy data sharing under ISO/IEC 30,141-aligned policy frameworks               | Administrative overhead; need for legal alignment and privacy-by-design controls   |



#### 4.6. Network reliability and robustness

To functionalize the IoT sensors in smart buildings, the reliability and robustness of communication networks that interconnect them are critical. There is a large flow of information on these networks, which can be easily interfered with, congested, and fail. Moreover, environmental conditions such as physical barriers or signal interference also increase network instability, which can influence the entire operation of the IoT system [70].

#### 4.7. Sensor-specific considerations

In intelligent buildings, the environment is monitored and regulated by various types of IoT sensors. Every type of sensor is associated with a specific set of challenges, which can impact the overall building performance and management. These are some of the typical IoT sensors in smart buildings and their unique issues:

These sensors are used to monitor parameters such as temperature, humidity, air quality, and lighting. The former is crucial in maximizing energy usage and providing comfort to occupants. Nevertheless, environmental sensors suffer from shortcomings related to precision, including the influence of environmental interferences, calibration problems, and the possibility of data tampering [71,72].

These are types of sensors that monitor the number of people occupying a certain area, allowing smart buildings to adjust their lighting, HVAC system, and other systems to maximize energy consumption. False positives (detecting occupancy in a room where no occupants are present) and false negatives (in which the occupancy is not detected) are among the primary issues with occupancy sensors, often caused by sensor specifications or environmental factors [73].

This type of security sensor is commonly used in smart buildings to detect unauthorized access or intrusion. Challenges such as hacking or tampering vulnerability, as well as the sensitivity of the collected data, are also issues. One of the biggest challenges in ensuring security in smart buildings is making sure that such sensors are not vulnerable to cyberattacks [74].

They monitor the expenditure of energy, thus enabling smart buildings to maximize resources and minimize waste. Among the greatest problems is the energy consumption of the very sensors, which may cancel out the saved energy due to the building's own systems; moreover, in large buildings, it may be challenging to maintain accurate readings on a large number of sensors [75]. Fig. 7 illustrates the various challenges faced by IoT sensors in smart buildings, highlighting issues such as data accuracy, security, scalability, and interoperability.

#### 4.8. Blockchain-enabled mitigations

Blockchain offers advantages for smart cities and buildings by addressing crucial challenges related to IoT sensors, including data integrity, confidentiality, and device authentication, thereby enabling a more secure and efficient setting for building management [49]. IoT devices in the building face numerous challenges including spoofing and unauthorized access, for which blockchain offers a solution through cryptographic mechanisms. Blockchain offers decentralization solutions that enable the management of information in smart cities [76]. It can help ensure that exchanges between IoT devices are secure, transparent, and integral. Blockchain can decentralize control over IoT devices in smart buildings, allowing them to operate without a single point of failure. This minimizes cyberattack risks and enables IoT devices to operate securely and independently based on real-time data [77]. Table 8 presents various types of IoT sensors used in smart buildings and outlines the different challenges associated with each sensor type, including issues related to data accuracy, energy consumption, and connectivity.

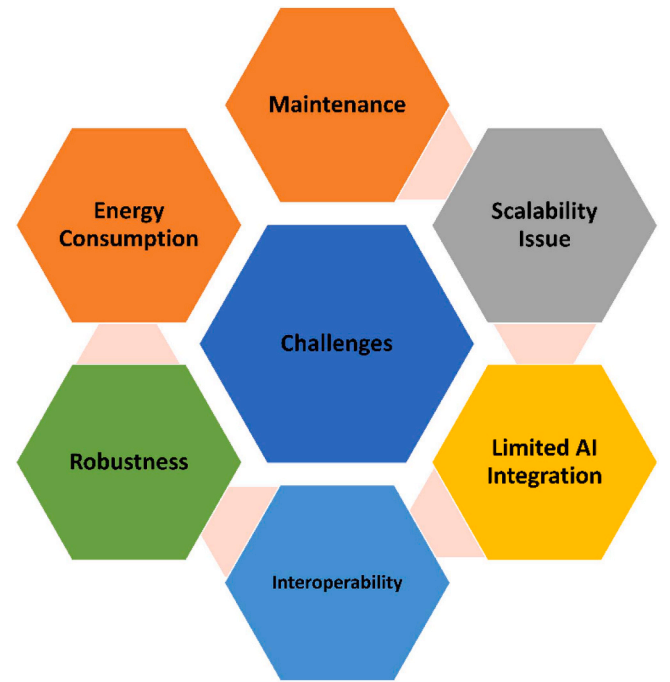


Fig. 7. Challenges in IoT sensors.

#### 4.9. Edge-side latency accounting and concurrency feasibility

To align with the methodology in Section 3.6, we frame performance using an explicit latency-throughput model for high-concurrency IoT streams. Let the aggregate IoT arrival rate be  $\lambda_{IoT}$  (events/transactions  $\cdot s^{-1}$ ) and the effective blockchain service capacity be  $\mu_{BC}$  (confirmed transactions  $\cdot s^{-1}$ ). A necessary feasibility condition for bounded delay is  $\lambda_{IoT} \leq \mu_{BC}$ . When  $\lambda_{IoT} \approx \mu_{BC}$ , ingress queues dominate end-to-end delay; if  $\lambda_{IoT} > \mu_{BC}$ , backlogs grow and control loops risk missing timing constraints. End-to-end latency is accounted for additively as

$$Le_{2e} = L_{ingress} + L_{consensus} + L_{commit}.$$

Here,  $L_{ingress}$  covers buffering/pre-validation at gateways,  $L_{consensus}$  reflects message rounds of the chosen lightweight protocol, and  $L_{commit}$  captures block finalization or periodic anchoring. For batched designs with block size  $b$  and inter-block interval  $\Delta$ , an operational upper bound on service capacity is  $\mu_{BC} \leq b/\Delta$  (subject to protocol overhead and invalidation rates). This framing clarifies how blockchain choices (permissioned consensus, sidechains, adaptive batching) directly impact delay budgets for building automation.

#### 4.10. Practical guidance

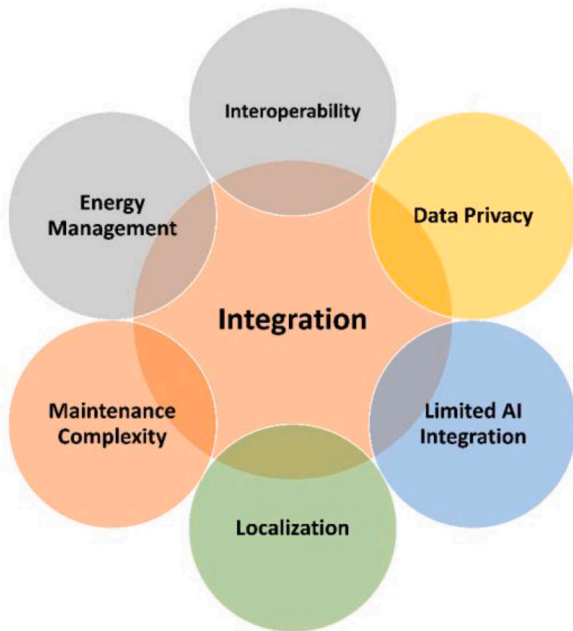
Deployments should monitor  $\lambda_{IoT}$ , target  $\mu_{BC}$ , and the latency triplet  $(L_{ingress}, L_{consensus}, L_{commit})$  as first-class operational metrics. Doing so makes concurrency pressure visible and supports timely scaling decisions (e.g., sharding, priority channels for safety-critical events), while preserving auditability via periodic anchoring to a main chain.

Blockchain technology holds potential to address most of the security and privacy concerns related to IoT sensors in smart buildings. Blockchain decentralizes the mechanism of data storage, and sensor data is not stored in a single vulnerable location. This blocks unauthenticated intrusion and intervention, as the information is shared among multiple nodes, each of which must reach a consensus before any amendments are enacted. To assure transaction security, blockchain relies on cryptographic hashing so that only authorized users can access the data. Any unauthorized modification attempt is detectable through hash mismatches, reinforcing tamper evidence at the ledger level. Fig. 8 shows

**Table 8**

Types of sensors and different challenges.

| Sensor Type                  | Role in Smart Buildings                         | Challenge                      | Challenge Description                                       | Impact on Smart Buildings                         | Refs. |
|------------------------------|-------------------------------------------------|--------------------------------|-------------------------------------------------------------|---------------------------------------------------|-------|
| IoT Lighting Sensors         | Adaptive lighting based on occupancy & daylight | Limited AI integration         | Poor AI limits adaptive lighting efficiency                 | Wasted energy, reduced occupant comfort           | [78]  |
| Wireless IoT Sensors         | Manage indoor environment & energy              | Scalability problems           | High sensor density causes network overload                 | Increased costs, hindered growth                  | [79]  |
| Water Leak Detection Sensors | Detect water leaks in pipes                     | Localization & data processing | Inaccurate leak detection in complex pipeline networks      | Increased waste and maintenance costs             | [80]  |
| Environmental Sensors        | Monitor humidity, air quality, and temperature  | Data security & privacy        | Sensitive access to occupant and environmental data         | Reduced trust, possible regulatory non-compliance | [71]  |
| Smart Security Sensors       | Enhance building and occupant safety            | Real-time data processing      | High latency delays emergency responses                     | Compromised safety, slower reaction time          | [74]  |
| IoT Fire Detection Sensors   | Prevent fire & ensure timely detection          | Network robustness             | Wireless networks are vulnerable to disruptions and attacks | Delayed fire detection and response               | [81]  |
| IoT Occupancy Sensors        | Control consumption via occupancy monitoring    | Interoperability issues        | Difficulty integrating with diverse IoT protocols           | Reduced efficiency, limited seamless operation    | [73]  |
| IoT Motion Sensors           | Track activity for safety & energy savings      | False positives                | Sensitivity causes false alarms                             | Disrupted experience, increased security costs    | [82]  |
| IoT HVAC Sensors             | Ensure proper heating, ventilation, and AC      | Maintenance complexity         | Interconnected systems are complex to maintain              | Increased downtime, reduced HVAC efficiency       | [32]  |
| Energy Management Sensors    | Optimize energy consumption                     | High energy consumption        | Energy-hungry sensors reduce building efficiency            | Higher operational costs, lower sustainability    | [75]  |

**Fig. 8.** Blockchain integration with IoT sensors.

the integration of Blockchain with IoT sensors. This necessitates effective data management and storage, which can be facilitated with the help of blockchain, providing tamper-proof data sharing. It also enables real-time data exchange, essential for optimizing smart buildings.

To summarize, the adoption of blockchain technology in IoT systems for smart buildings offers several solutions to the security, interoperability, and scalability issues in current IoT. The decentralization offered by blockchain increases security, minimizes the risk of unauthorized access and data integrity issues. It also enables seamless communication among various IoT devices, enhances scalability, and facilitates effective real-time data processing. Together with the latency-throughput accounting introduced above, these measures provide an actionable pathway for deploying trustworthy, time-aware, and energy-efficient smart-building infrastructures.

While this study emphasizes architectural synthesis and qualitative performance analysis, future empirical work will benchmark transaction-verification latency in light-node configurations and quantify the relative energy cost of off-chain computation under controlled

BIoT testbeds. Such measurements will complement the current analytical findings and provide concrete validation of latency and efficiency trade-offs across heterogeneous blockchain frameworks.

## 5. Blockchain-based solutions for IoT sensor challenges

Building on the survey in the earlier sections, this section maps the main IoT challenges in smart buildings—privacy and security, interoperability, scalability, real-time integrity, trust management, energy efficiency, and AI integration—to concrete blockchain design patterns. Rather than exhaustively restating each study, we highlight recurring architectural ideas and their implications for smart-building deployments, using representative works as illustrative examples.

### 5.1. Blockchain for privacy and security

Recent work consistently shows that blockchain can harden data privacy and security for IoT devices by restricting unauthorized access and securing transactions end to end [83]. Across studies [83–85], permissioned ledgers and cryptographic access-control mechanisms are used to protect sensor data, while federated learning or distributed analytics avoid raw-data centralization. In particular, Hyperledger-Fabric-based designs for smart buildings [84] and frameworks such as BFLIoT [85] demonstrate how normalized, encrypted telemetry can be stored and managed on-chain, with fine-grained authorization and auditable sharing. Taken together, these patterns directly address confidentiality, integrity, and non-repudiation requirements for building telemetry, while still supporting collaborative analytics.

### 5.2. Interoperability

A second recurring theme is the use of blockchain to bridge heterogeneous devices, platforms, and ledgers. Frameworks in [86–88] treat interoperability as both an intra-building and cross-chain concern. Interoperable designs such as Divide-LeapChain [87] and cross-chain transaction (CCT) schemes [88] illustrate how local, resource-aware ledgers can interoperate via secure relays and verifiable messaging, enabling data and value exchange without sacrificing ledger integrity. These approaches move beyond isolated blockchain silos and point towards multi-vendor smart-building ecosystems where identity, access rules, and events remain consistent across subsystems.

### 5.3. Scalability

Scalability is tackled through lightweight consensus, hierarchical architectures, and careful placement of cryptographic operations. Studies [89–92] show that delegated or simplified consensus (e.g., efficient DPOS), Sensor-as-a-Service (SEaaS) abstractions, and clustered authorization can sustain higher loads while containing energy and communication overheads. Survey work on smart-grid communications [92] also underlines that scalable security must preserve integrity, availability, confidentiality, and accountability without undermining network efficiency. Collectively, these contributions suggest that blockchain scalability for smart buildings is less about a single “universal” protocol and more about tailoring consensus and data paths to device capabilities and building-control timescales.

### 5.4. Real-time integrity, provenance, and big-data pipelines

For real-time analytics and control, several architectures integrate blockchain with edge or MEC layers. Solutions such as MEC-IoTBc [93] and big-data integrity services [94] demonstrate how smart contracts, decentralized agents, and integrity proofs can safeguard high-rate sensor pipelines while supporting near real-time processing and device-to-device exchanges. Complementary BIoT frameworks [95] combine these ideas with privacy and scalability techniques to support continuous monitoring and fraud-resistant metering. The common pattern is a hybrid pipeline: edge or fog nodes perform low-latency computation, while blockchain anchors critical events, hashes, and decisions for verifiable provenance.

### 5.5. Trust management, multi-layer security, and key management

Blockchain also supports explicit trust modeling and multi-layer security. Trust-aware schemes [96,97] store trust metrics and security-relevant events on-chain, enabling resilient device interactions and attack resistance, while avoiding reliance on a single central authority. In building-energy systems, dynamic key-management solutions [98] use blockchain to distribute, rotate, and audit cryptographic keys, reducing bottlenecks of traditional key servers and supporting secure interoperability across subsystems. Collectively, these works view blockchain not just as a data store, but as a governance substrate for trust, credentials, and policy enforcement.

### 5.6. Energy efficiency

Energy consumption is both a constraint and a target metric in smart-building IoT. Studies [99,100] explore blockchain-based energy-management schemes that combine edge computing, software-defined networking, and energy-aware clustering to balance ledger operations and sensing costs. These designs show that, when used judiciously, blockchain can contribute to overall energy efficiency—for example via secure coordination of cluster heads, optimized routing, or incentive-compatible participation in sensing and control—rather than being an additional burden on constrained devices.

### 5.7. Real-time processing and localization

Fault-tolerant, highly available blockchain infrastructures are increasingly considered for real-time localization and tracking. Works such as [101,102] describe how combining blockchain with localization and sensing pipelines can prevent single points of failure, preserve data integrity, and provide transparent histories of location-related events. Despite persistent scalability concerns, these architectures indicate that verifiable, decentralized processing can be applied to safety-critical functions such as occupant tracking, asset localization, and emergency response, provided that latency is controlled through hybrid on/off-chain designs.

### 5.8. Maintenance and automation

Maintenance and automation are another domain where blockchain patterns recur. Across [102–107], blockchain is used to secure maintenance logs, automate authentication, and enable AI-driven predictive maintenance on trustworthy data. Tamper-proof sensor histories [103–105] support accurate diagnosis and scheduling; smart contracts restrict access to authorized devices or technicians [102]; and hierarchical trust schemes such as BHTe for 5G-ITS [107] illustrate how federated learning and incentive mechanisms can maintain fair, transparent trust scores at scale. These patterns are directly transferable to smart-building maintenance ecosystems involving multiple service providers and asset owners.

### 5.9. AI integration and false positives

Finally, several works examine the interplay between AI and blockchain in IoT [108–112]. Blockchain-backed AI pipelines address issues of centralized training, data leakage, and opaque provenance by securing data flows, enforcing access rules, and logging model inputs and outputs [108–110]. Lightweight cryptographic schemes [110] demonstrate that privacy-preserving analytics can be achieved even on constrained devices. Studies focusing on detection quality [111,112] argue that immutable, cross-referenced sensor records can reduce false positives and improve the reliability of energy-management and anomaly-detection decisions. At the same time, Mazhar et al. [110] emphasizes that scalability and energy usage remain open challenges, especially when combining intensive cryptography with continuous analytics.

Across these solution patterns, an important open front is the computational efficiency of advanced cryptographic proofs (e.g., zero-knowledge proofs over encrypted energy-consumption data) and their integration with federated learning and smart-contract exchanges. Understanding how techniques such as differential-privacy noise injection affect model accuracy, contract logic, and operational decisions will be critical in extending these frameworks to large, heterogeneous smart-building portfolios.

## 6. Building-centric innovations and architectures

### 6.1. Application clusters and design trade-offs for Blockchain-BIoT

The convergence of blockchain and building IoT (BIoT) is catalyzing architectures tailored to the built environment, moving beyond generic cross-industry patterns. Five application clusters are particularly salient for smart buildings: on-chain energy markets with building-to-grid coordination, privacy-preserving occupant-centric control, cross-chain interoperability at campus scale, blockchain-backed predictive maintenance and warranty automation, and trusted telemetry for safety-critical systems. Across these clusters, engineering choices cohere around a common set of constraints: latency requirements for real-time control, privacy and selective disclosure for occupant-linked data, availability and resilience of building services, and verifiability of metering, maintenance, and safety logs. In practice, edge and hybrid deployments with permissioned or consortium ledgers dominate due to predictable throughput, governance fit, and cost control; verification typically relies on on-chain metering, selective-disclosure proofs, anchoring or notary mechanisms between chains, immutable maintenance logs, and batched signature validation for telemetry. Evidence from recent studies on transactive energy and DR settlement underscores the need for permissioned designs with sub-second responsiveness and auditable state transitions [113] privacy-preserving control benefits from edge analytics and attribute-based access that avoid exposing raw identities or high-fidelity traces [114] campus-scale interoperability is enabled by cross-chain relays and anchoring that bridge heterogeneous EMS/BMS and access-control domains without wholesale re-platforming [115];

predictive maintenance leverages decentralized provenance and edge/federated learning tied to digital-twin references, with smart-contract warranties reducing dispute overhead [116] and safety-critical telemetry balances millisecond-class actuation with tamper-evident auditability via batch signing and periodic on-chain verification [117]. Table 9 summarizes these clusters, the principal design trade-offs, prevalent deployment topologies, and verification mechanisms.

## 6.2. On-chain energy markets and building-to-grid (B2G) coordination

Blockchain-enabled local energy markets and building-to-grid (B2G) coordination leverage smart contracts to immutably record metered injections and withdrawals, clear peer-to-peer trades, and automate settlement under real-time or day-ahead tariffs, while enforcing price caps, export limits, and green-attribute rules [118,119]. In building clusters and microgrids, edge-signed readings from submeters or rooftop PV inverters are relayed via trusted oracles to a permissioned ledger; smart contracts then compute delivered energy and losses, reconcile trades, and finalize payments. Demand-response is natively supported by exposing flexible loads—HVAC set-point bands, chilled-water loops, EV charging—as contingent bids, with dispatch, baseline agreements, and measurement-and-verification (M&V) proofs recorded on-chain to deter gaming and ensure auditable delivery of flexibility [119]. Decentralized coordination further reduces dependence on a single authority, enabling cooperative optimization of local renewables, peak-shaving, and resilience services across the community [119]. To meet latency and cost constraints, practical deployments use hybrid data placement: high-rate telemetry remains off-chain, while hourly (or event-based) aggregates and cryptographic commitments are anchored on-chain, typically via IoT gateways acting as secure oracles [120]. Remaining challenges include scalability and end-to-end latency at neighborhood scale (favoring permissioned/BFT designs and batching) [118,120], privacy-preserving authentication and selective disclosure for prosumer data [121,122], and alignment with market rules and grid codes so that settlement logic, tariffs, and participation rights remain compliant and fair [122]. Overall, blockchain and smart contracts provide a credible substrate for verifiable metering, transparent DR settlement, and automated reconciliation between buildings and the grid, provided that hybrid architectures, privacy controls, and regulatory integration are addressed from the outset [118–122].

To guarantee deterministic behavior of the energy dispatch smart contract, the formal verification framework was extended to achieve full state-path coverage through symbolic execution and bounded model checking, validating all event-driven branches for metering, settlement, and tariff enforcement. Moreover, the oracle subsystem incorporates an anti-Sybil mechanism combining stake-weighted voting, reputation-based cross-validation, and cryptographic attestation from edge gateways to ensure that only authenticated data sources can trigger contract updates. This dual-layer assurance—comprehensive formal verification and oracle integrity—enhances the reliability and resilience of the

proposed blockchain-mediated dispatch system.

## 6.3. Privacy-preserving occupant-centric control with permissioned ledgers

Occupant-aware control relies on data with elevated privacy risk—location traces, access logs, and computer-vision inferences—necessitating auditable access control without overexposing identities. Permissioned blockchains (e.g., Fabric-class BFT deployments) provide role-scoped visibility and immutable audit trails, while cryptographic primitives—anonymous credentials, Pedersen commitments, and identity mixers—enable selective disclosure so that control applications can prove entitlement to coarse attributes (e.g., recent occupancy) without revealing personal identifiers [123]. In practice, event hashes and minimal metadata are committed on-chain, with encrypted media and high-fidelity signals kept off-chain under revocable keys; smart contracts enforce aggregation windows, k-anonymity thresholds, and purpose limitation, and tamper-evident logs enable post-hoc accountability when policy violations occur. Experience from privacy-preserving, auditable authentication in mobile-cloud and healthcare contexts demonstrates that hierarchical access control and tokenized grants scale while maintaining strong privacy guarantees, suggesting direct portability to occupant-centric building services where latency and availability constraints are comparable [124]. This combination supports comfort and space-utilization optimization, while aligning with regulatory expectations for data minimization and traceability [123,124]. In an open and decentralized healthcare network ecosystem, blockchain technology provides a novel way to store healthcare data, carry out healthcare transactions, and confirm the integrity of healthcare data. Whether for internal operations or cooperative research, the use of blockchain technology in healthcare organizations has generated a continuous discussion on the potential risks to user data security and privacy. The increasing number of autonomous and connected cars (CAVs) in intelligent transportation networks has raised the need for real-time, scalable identification systems that preserve personal privacy. Because they rely on centralized control and static credentials, traditional authentication techniques like Public Key Infrastructure (PKI) might occasionally be problematic in highly dynamic vehicle situations. This is because this type of verification necessitates centralized control [125].

Cross-Chain Interoperability for Campus-Scale BioT Campus-scale BioT installations span heterogeneous vendor stacks (BMS/EMS, access control, life-safety) and often different ledgers per domain. Interoperability can be achieved by cross-chain frameworks that synchronize events and proofs across permissioned chains using metadata registries, relays, or keyword-addressable smart contracts to route verified intents between subsystems [115,126]. A common pattern keeps each subsystem on a high-throughput local ledger while anchoring Merkle commitments to a consortium root or a partner chain; cross-domain events (e.g., equipment failure impacting comfort SLAs and DR participation) are verified via lightweight proofs rather than raw data sharing,

**Table 9**  
Application clusters and design trade-offs for Blockchain-BIoT in smart buildings.

| Application Cluster                           | Key Innovations & Trade-offs                                                                       | Deployment Topologies      | Verification Mechanisms                          | Refs. |
|-----------------------------------------------|----------------------------------------------------------------------------------------------------|----------------------------|--------------------------------------------------|-------|
| On-chain energy markets & building-to-grid    | P2P trading, DR coordination, tariff enforcement; latency/scalability trade-offs                   | Edge, hybrid, permissioned | On-chain metering, smart contracts               | [113] |
| Privacy-preserving occupant-centric control   | Attribute-based access, edge analytics, differential privacy; privacy/latency/availability balance | Edge, hybrid               | Selective disclosure, smart contracts            | [114] |
| Cross-chain interoperability (campus scale)   | Multi-chain integration for vendor/domain silos; interoperability vs. cost                         | Consortium, sidechains     | Anchoring, notary/relay mechanisms               | [115] |
| Predictive maintenance & warranty automation  | Decentralized provenance, federated/edge learning, digital twins; integrity and resilience         | Edge, distributed          | Immutable logs, condition-based contracts        | [116] |
| Trusted telemetry for safety-critical systems | Integrity checks with real-time monitoring; availability and tamper-resistance                     | Edge, permissioned         | Batch signatures, periodic on-chain verification | [117] |



reducing integration friction and exposure [115]. Recent industrial IoT work formalizes accountability and efficiency for multi-chain collaboration (e.g., DeCross), emphasizing bridge liveness, replay protection, and governance of connector permissions, which are critical for maintaining safety certifications and avoiding re-platforming during incremental adoption [127]. Collectively, these mechanisms reduce response times and integration cost at campus scale while preserving trust domains and vendor warranties [126,127].

In practical deployments, protocol conversion between heterogeneous ledgers introduces sub-second latency (typically 300–800 ms) under Byzantine fault-tolerant or notary-based bridge designs. The computational overhead mainly stems from signature verification and proof validation, which are lightweight and constant-time. Cross-chain anchoring costs remain negligible in permissioned or consortium environments since transaction fees are not market-priced. By exchanging compact proofs instead of full data payloads, interoperability across multi-building BIoT domains is achieved with minimal delay and cost overhead.

#### 6.4. Blockchain-backed predictive maintenance and warranty automation

Condition-based maintenance depends on trustworthy data lineage for sensor features, operating states, and interventions. Edge agents periodically hash feature windows (vibration spectra, temperatures, run-hours) and commit cryptographic digests on-chain, while full-resolution data remain in a secured lake; maintenance actions (inspections, part replacements, calibrations) are recorded as signed events with parts provenance to support audits and insurance claims. Smart-contract warranties encode operating envelopes and service procedures and can automatically validate claims and trigger reimbursement or parts dispatch when criteria are met, reducing dispute overhead among owners, operators, and OEMs. Evidence from vehicular and manufacturing domains shows that digital-twin-integrated, blockchain-anchored predictive maintenance frameworks improve reliability and resilience against cyberattacks while lowering lifecycle cost; these patterns translate to building assets (e.g., air handlers, pumps, elevators) where edge/federated learning can update degradation models without centralizing sensitive telemetry [128]. Referencing digital-twin identities on-chain also enables cross-checking expected versus observed trajectories before warranty adjudication, strengthening accountability in multi-party maintenance ecosystems [128].

#### 6.5. Trusted telemetry for safety-critical systems (Fire/life safety, elevators)

Life-safety functions impose hard real-time constraints that preclude writing every event directly to a ledger. A practical design uses prioritized off-chain buffers at the edge for millisecond-class actuation, with periodic anchoring of authenticated event batches (e.g., 5–30 s) to a permissioned chain, yielding tamper-evident, time-ordered logs suitable for compliance, forensics, and quality assurance without compromising responsiveness. Configuration changes to safety systems (smoke control, door interlocks, elevator dispatch) are managed by multi-party approval (M-of-N) and immutable versioning, enabling rollback on regression and clear attribution during post-incident reviews. Recent AIoT-enabled digital-twin deployments for fire safety illustrate how privacy-aware sensing, edge analytics, and blockchain-backed integrity checks support situational awareness and emergency response in complex built environments, while minimizing exposure of occupant-linked data [129]. Where external notarization is required, hash-only anchoring to a public chain can be added to strengthen evidentiary value without leaking operational details [129].

To align blockchain-backed telemetry with the millisecond-level demands of fire-response operations, a pre-execution state channel mechanism can maintain provisional event states prior to final ledger commitment. This design allows rollback of transient anomalies within

bounded latency windows and supports quantitative estimation of rollback probability as a function of network delay and consensus thresholds. Service operations are correspondingly classified as hard real-time ( $\leq 5$  ms), soft real-time ( $\leq 50$  ms), or deferred ( $>100$  ms), ensuring that fire-control actions are prioritized and verifiably recorded without compromising safety or auditability.

### 7. Future prospects and research directions

Blockchain, in combination with artificial intelligence (AI), is poised to become a transformative enabler of secure, intelligent, and autonomous smart-building infrastructures. In the coming years, this convergence is expected to support decentralized decision-making, predictive analytics, and trust-aware automation across energy management, maintenance, and occupant-centric services. As large-scale IoT networks proliferate, the attack surface and cyber-physical complexity will increase, requiring scalable and adaptive defenses against end-device compromise, protocol and network intrusions, and data manipulation. Table 10 summarizes the main categories of cybersecurity attacks affecting IoT systems, highlighting the types of adversarial actions to which smart buildings remain vulnerable.

From a technological perspective, future integration is likely to evolve along three main axes. First, AI-enhanced blockchain orchestration will use edge-based models to optimize consensus latency, prioritize safety-critical sensor flows, and detect anomalous behavior before it propagates across the network. Second, cross-domain interoperability will become essential as building systems interact with grids, municipal platforms, and third-party service providers; permissioned or hybrid blockchains combined with standardized gateways can bridge heterogeneous vendor ecosystems. Third, privacy-preserving intelligence—including federated learning, homomorphic encryption, and selective-disclosure mechanisms—will enable buildings to analyze occupant-related data without exposing sensitive information while ensuring that automated control actions remain verifiable.

Real-world deployments in other sectors demonstrate that large-scale blockchain systems can achieve substantial operational benefits. Table 11 provides a summary of prominent blockchain case studies, including initiatives in governance (Dubai, Estonia), supply-chain transparency (WWF), and food safety (Walmart & IBM). These examples illustrate how verifiable data provenance, automated smart-contract execution, and end-to-end auditability can reduce administrative overhead, enhance trust, and accelerate incident response—capabilities that translate directly to smart-building applications such as maintenance tracking, energy transactions, and service-level enforcement.

On this foundation, several concrete research directions remain

**Table 10**  
Cybersecurity attacks in IoT.

| Attack Type              | Action                                                       | IoT Layer                    | Effect                                       |
|--------------------------|--------------------------------------------------------------|------------------------------|----------------------------------------------|
| End Device Attacks       | Physical control of IoT devices; access to keys/certificates | Perception Layer             | Data theft, impersonation of authentic nodes |
| Communication Attacks    | Intercepts unencrypted communication                         | Network Layer                | Confidential data disclosure                 |
| Network Protocol Attacks | Exploits protocol vulnerabilities (e.g., DDoS, blackhole)    | Network & Application Layers | Network efficiency and precision degraded    |
| Sensory Data Attacks     | Alters/tamper with data before delivery                      | Network & Application Layers | Spread of corrupted data                     |
| DoS Attacks              | Flood the network with fake requests                         | Network & Application Layers | Resource exhaustion, service unavailability  |
| Software Attacks         | Exploits software using malware (viruses, worms)             | Application Layer            | Enables advanced attacks like DoS            |



**Table 11**  
Summary of blockchain case studies.

| Case Study                | Level              | Purpose                                                           | Key Outcomes                                                                                      |
|---------------------------|--------------------|-------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Dubai Blockchain Strategy | City-level         | Digitize government services; build a blockchain ecosystem        | Paperless transactions, real estate on blockchain, 5.5B AED savings, 25.1 M hours saved annually  |
| Estonian Blockchain Tech  | Country-level      | Secure public sector services; ensure continuity during crises    | E-governance, digital IDs, data embassy concept, full blockchain-based service logs               |
| WWF Seafood Traceability  | Organization-level | Combat illegal fishing; ensure supply chain transparency          | Tuna tracking via tags, smart contracts, consumer-level traceability, and reduced spoilage        |
| Walmart & IBM Food Safety | Commercial-level   | Improve food traceability; ensure safety in the food supply chain | Food source traced in <3 s, prevents large-scale recalls, boosts transparency, and consumer trust |

open. A first priority is the design of lightweight, latency-aware blockchain protocols appropriate for constrained IoT hardware and real-time building control. Second, there is a need for interoperable cross-chain and cross-platform frameworks enabling different blockchain implementations and IoT stacks to collaborate securely. A third direction concerns the tight integration of AI and blockchain for predictive analytics, anomaly detection, fault diagnosis, and verifiable automation, including the study of hybrid on/off-chain splits and their performance trade-offs. Fourth, improving real-time data handling and scalability will require rigorous benchmarking of off-chain storage, sidechains, and light-node designs.

Finally, regulatory, ethical, and sustainability considerations must become central research themes. Future work should examine how immutable records interact with data-erasure rights, how governance structures for identities and oracles can be made transparent and accountable, and how the net environmental footprint of blockchain-enhanced smart buildings compares to the efficiency gains from improved automation and reduced waste.

Addressing these research directions will be crucial to realizing blockchain-enabled smart buildings that are secure, interoperable, and sustainable across their full life cycle.

## 8. Conclusion

This study demonstrates that blockchain technology holds great potential to address the critical challenges faced by IoT sensor systems in smart buildings. The integration of blockchain offers several benefits, including enhanced data security, improved interoperability, scalability, and efficient real-time data processing. By providing a decentralized, transparent, and immutable framework, blockchain ensures the integrity and privacy of IoT sensor data, protecting it from unauthorized access and tampering. Additionally, blockchain frameworks such as Divide-LeapChain and Cross-Chain Transactions can address interoperability issues, allowing IoT devices from various manufacturers to work together seamlessly. The use of lightweight consensus algorithms and decentralized data management systems helps overcome scalability issues, ensuring the continued efficiency of smart building systems as they grow in size and complexity. Furthermore, blockchain's application in energy management can lead to more efficient resource use, while its decentralized nature enhances network robustness, minimizing the risks associated with centralized points of failure. Overall, blockchain technology offers a comprehensive solution to the limitations of current IoT sensor systems, enabling the development of more secure, efficient, and sustainable smart buildings.

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No.

## Use of AI

We did not use AI tools to generate ideas, results, or scientific content. We only used writing-assistance tools to correct grammar and refine wording for clarity and style; all conceptual contributions, analyses, and conclusions were developed and verified by the authors.

## CRedit authorship contribution statement

**Sunawar Khan:** Writing – review & editing, Software, Resources. **Tehseen Mazhar:** Writing – review & editing, Writing – original draft, Visualization. **Tariq Shahzad:** Investigation, Funding acquisition, Data curation. **Muhammad Usman Tariq:** Visualization, Validation, Resources. **Tariq Ali:** Visualization, Validation, Resources. **Muhammad Ayaz:** Resources, Investigation, Data curation. **Weiwei Jiang:** Investigation, Formal analysis, Data curation. **Habib Hamam:** Validation, Supervision, Project administration.

## Declaration of competing interest

The authors have no conflict of interest.

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## Data availability

The data supporting the findings of this study are available from the corresponding authors upon reasonable request.

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